

✓ Homework 6: Feature Engineering

After going through the lecture notebooks, work on the homework below.

Question

1. Given the MyAnimeList dataset [1], **Standardize** the `Score` column (mean=0, std=1).
2. Apply **One-Hot Encoding** to the `Type` column and convert to `int` (0/1).
3. Return a `pd.DataFrame` combining the standardized `Score` and the one-hot encoded `Type` columns (`Movie`, `Music`, `ONA`, `OVA`, `Special` and `TV`).

The resulting DataFrame should have the following columns (in this order):

1. `Score_std`
2. One-hot encoded `Type` columns (`Movie`, `Music`, `ONA`, `OVA`, `Special` and `TV`).

Inputs / Outputs:

```
homework(anime_data_extracted: pd.DataFrame) -> pd.DataFrame
```

Argument	Type	Description
<code>anime_data_extracted</code>	<code>pd.DataFrame</code>	Preprocessed anime dataset (see Section 2)
Return	Type	Description
<code>result</code>	<code>pd.DataFrame</code>	DataFrame with <code>Score_std</code> as the first column, followed by one-hot encoded <code>Type</code> col

Example:

```
result = homework(anime_data_extracted)
print(result.head())
```

```

   Score_std  Movie  Music  ONA  OVA  Special  TV
0  2.5XXXXX     0     0     0     0         0    1
1  2.1XXXXX     1     0     0     0         0    0
2  1.9XXXXX     0     0     0     0         0    1
3  0.8XXXXX     0     0     0     0         0    1
4  0.5XXXXX     0     0     0     0         0    1

```

Submission Guidelines:

1. Submit the entire `homework()` function by selecting this week's assignment in the Omnicampus homework section, pasting the function into the submission area, and then clicking [Submit Python Code].

Note:

- Erase the `!!WRITE ME!!` when submitting
- The return value of the function should be type `pd.DataFrame`
- Write your answer as one function
- You may use any Generative AI (NotebookLM recommended)

Deadline

Wed, Jun 10 (11:00 UTC)

✓ 1. Importing Libraries

Note: Do not include this section in your `homework()` function.

```
import numpy as np
import pandas as pd
from pandas import DataFrame

# Libraries for retrieving data from the web and handling zip files
import requests, zipfile
from io import StringIO
import io
```

✓ 2. Downloading the Dataset

Download the MyAnimeList dataset from GitHub and clean the data by removing rows where `Score` is `'Unknown'`.

Note: Do not include this section in your `homework()` function.

```
# Specify the url with data
url = 'https://github.com/Hernan4444/MyAnimeList-Database/archive/refs/heads/main.zip'

# Acquire data from the url
```

```

r = requests.get(url, stream=True)

# read and extract the zipfile
z = zipfile.ZipFile(io.BytesIO(r.content))
z.extractall()

# Load and clean the data
anime_data = pd.read_csv('MyAnimeList-Database-master/data/anime.csv')
anime_data_extracted = anime_data[anime_data['Score'] != 'Unknown'].copy()
anime_data_extracted['Score'] = pd.to_numeric(anime_data_extracted['Score'])

```

✓ 3. Solution

Write your solution in the `homework()` function below. The function receives `anime_data_extracted` as input and should return a `pd.DataFrame`.

```

# Fill in the WRITE ME
# Don't forget to erase the "!!WRITE ME!!" before submitting!
def homework(anime_data_extracted):
    !!WRITE ME!!

```

✓ 4. Test Your Solution

Run the test cell below to check that your `homework()` function does not contain import statements or data loading code before submitting.

Note: Do not modify the test code.

```

import ast
import inspect
import textwrap

def hw7_public_tests(homework_func):
    print("=== HW6 Public Tests ===")

    # Get the source code of the homework function
    source = textwrap.dedent(inspect.getsource(homework_func))
    tree = ast.parse(source)

    # ----- Test 1: No import statements -----
    print("Test 1: No import statements in function...", end=" ")
    for node in ast.walk(tree):
        if isinstance(node, (ast.Import, ast.ImportFrom)):
            print("NG")
        if isinstance(node, ast.Import):

```

```

        names = ", ".join(alias.name for alias in node.names)
    else:
        names = node.module
    print(f" Found import statement: '{names}'")
    print(" Remove all import statements before submitting.")
    return
print("OK")

# ----- Test 2: No data loading code -----
print("Test 2: No data loading code in function...", end=" ")
blocked_calls = {"read_csv", "read_excel", "read_json", "read_html",
                  "get", "post", "ZipFile", "extractall", "urlopen"}
for node in ast.walk(tree):
    if isinstance(node, ast.Call):
        func_name = ""
        if isinstance(node.func, ast.Attribute):
            func_name = node.func.attr
        elif isinstance(node.func, ast.Name):
            func_name = node.func.id
        if func_name in blocked_calls:
            print("NG")
            print(f" Found data loading call: '{func_name}()'")
            print(" Do not include data downloading/loading code in your f
            return
print("OK")

# ----- Test 3: No hardcoded file paths or URLs -----
print("Test 3: No hardcoded file paths or URLs...", end=" ")
for node in ast.walk(tree):
    if isinstance(node, ast.Constant) and isinstance(node.value, str):
        val = node.value
        if val.startswith("http") or ".csv" in val or ".zip" in val:
            print("NG")
            print(f" Found hardcoded path/URL: '{val[:60]}...'")
            print(" Do not include file paths or URLs in your function.")
            return
print("OK")

print("=== All Public Tests Passed ===")

hw7_public_tests(homework)

```

Once all tests have passed, copy and paste your `homework()` function to Omnicampus and submit your solution. If you see a `1.0`, it means that your answer was correct.

✓ References

[1] Hernan4444. *MyAnimeList Database*. GitHub.

<https://github.com/Hernan4444/MyAnimeList-Database>

Double-click (or enter) to edit